CSE – II

504 – 19
505 – 31
507 – 14

64

(Please Write your roll No. Immediately) Roll No.:

First Term Examination, Februray-2019

B.Tech. (VI sem) CSE/IT

Paper Code : ETCS-302

Subject: Compiler Design

Time : 1.5 hours

Max. Marks:30

Note: Attempt Q. No. 1 which is compulsory and any two other questions from remaining questions. Each question carries 10 marks.

Q.1 a) How dependency graph effects the evaluation order in syntax directed translation?

b) "Lexical Analysis is the most time consuming phase in compilation" justify this statement.

c) Create NDFA for the regular expression $(a+b)^*abb+bba$.

d) Differentiate between L-Attributed definition and S-Attributed definition.

e) Remove left recursion from the following grammar: S→AaAb|b A→Ab | a | ab [2*5=10]

Q.2. Prove that following grammar is CLR(1) but not LALR(1)

S→Aa |bAc|Bc|bBa

A→d

B→d

[10]

Q.3.

(a) Calculate first and follow for all non-terminals for the following grammar.

E→E+T|T

T→TF|F

F→F*|a|b

[6]

(b) Create a predictive parsing table for the grammar shown in part a. [4]

Q.4.

(a) How lex(flex) tool can be used to create lexical analyzer? Explain with the help of an example.

[5]

(b) Define Synthesized and inherited attributes. Identify Synthesized and inherited attributes from the grammar given below:

D→TL	L.in=T.type; L.x=T.y
L→L_1,id	L_1.in=L.in; L.x=L_1.x
T→id	

Justify your answer. [5]

(Please Write your roll No. Immediately) Roll No.:

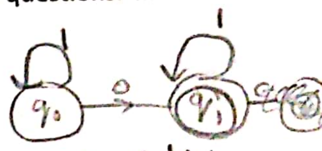
First Term Examination, Februray-2018

B.Tech. (VI sem) CSE/IT

Paper Code: ETCS-302 Subject: Compiler Design

Time: 1.5 hours Max. Marks:30

Note: Attempt Q. No. 1 which is compulsory and any two other questions from remaining questions. Each question carries 10 marks.



Q.1 a) List the various phases of compilation process.

b) Generate a pattern matcher which can recognize following two patterns: 01* and 1*0.

c) Improve the grammar by removing left recursion. S→AaB | b A→Aab | Aba|a B→a.

S → AaB | b
A → Aab | Aba | a
B → a

d) Differentiate between Synthesized attributes and inherited attributes.

e) Define the process used by lex tool for generating lexical analyzer. [2*5=10]

Q.2. Consider the grammar given below ε =NULL

X→0Y1 |1Z2

Y→2 | ε

Z→1Y2 | 1| ε

	First	Follow
X	{0, 1}	{$}
Y	{2, ε}	{1, 2}
Z	{1, ε}	{2}

(a) Calculate first and follow for the non-terminals. [5]

(b) Create a predictive parsing table for the above grammar. Also Show that this grammar is not LL(1). [5]

Q.3. Consider the grammar

S→L=R | R

L→*R | id =|$

R→L

	First	Follow
S	{*, id}	{$}
L	{*, id}	{=, $}
R	{*, id}	{$, =}

Design LALR parser for this grammar. [10]

Q.4.

(a) Describe the functioning of YACC with the help of an example. [5]

(b)Define following terms:

(i) Topological Sort (ii) Dependency Graph (iii) Attribute (iv) Circular SDT

(v) Evaluation order [5]

$S \to AaB \mid b$

$A \to A\underbrace{ab}_{\alpha} \mid \underbrace{a}_{\beta} \Rightarrow A \to aA'$
$\qquad A' \to abA' \mid \epsilon$

$A \to A\underbrace{ba}_{\alpha} \mid \underbrace{a}_{\beta}$
$\qquad A' \to aA'$
$\qquad A' \to baA' \mid \epsilon$

$B \to a$

(C)

$$\boxed{\begin{array}{l} S \to AaB \mid b \\ A \to aA' \\ A' \to abA' \mid baA' \mid \epsilon \\ B \to a \end{array}}$$

Ques 2 Predictive parsing table :-

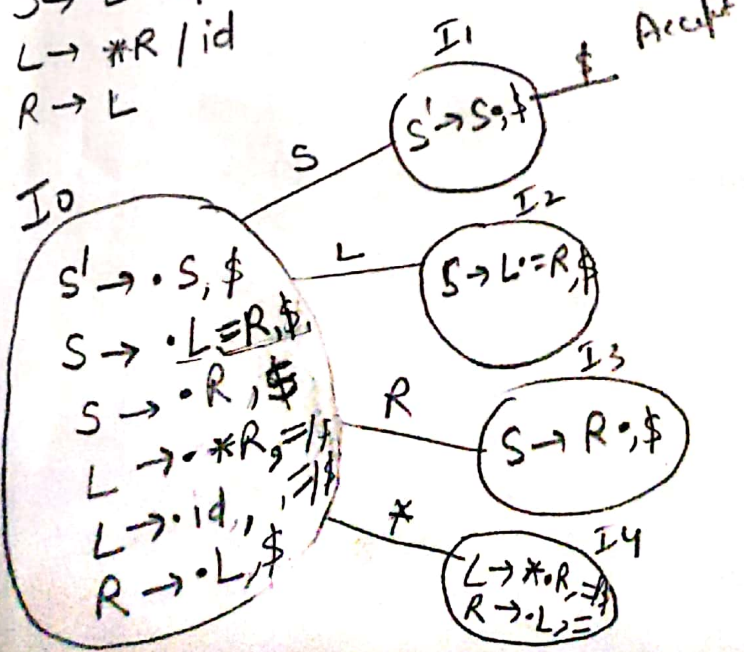
LL(1) Parsing table

	0	1	2	$	=
X	X→0Y1	X→1Z2			
Y	~~Y~~	Y→ε	Y→2 Y→ε		
Z		Z→1Y2 Z→1	Z→ε		

So the grammar is not LL(1)

Ques 3 :- LALR Parser :-

$S \to L=R \mid R$
$L \to *R \mid id$
$R \to L$



(Please write your Roll Number) CSE-I

~~404 — 24~~
405 — 32
~~407 — 10~~
66

First-Term Examination

Sixth Semester (B.Tech) Feb-2016

Paper Code: ETCS-302 Subject: Compiler Design

Time: 1 Hr 30 Mins Max. Marks: 30

Note: Q1 is compulsory and attempt any two questions out of remaining three.

Q1 Attempt any five parts. (5*2=10)

a) Discuss merits and demerits of single pass compiler and multipass compiler.

b) What do you mean by bootstrapping in compilation?

c) What is the need for separating the parser from scanner?

d) Find leftmost derivation, rightmost derivation and derivation tree of a string 'aacbbcc' from following grammar productions {S→aAcB|Bds , B→aAcA|cAB|b , A→aB|aBc|a}.

e) Explain handle pruning with an example.

f) What are the responsibilities of Loader, Linker and Assembler in the compiler environment?

g) Write steps involved in compilation of following sentence

pos = interest + rate * 60

Q2 Attempt both parts (2*5=10)

a) Remove left recursion from the following grammar:

S → Aa | b

A→Ac | Sd | e

A → Ac | Aad | bd | e

S → Aa | b
A → bdA' | eA'
A' → AcA' | AadA' | ε

b) Prove whether following grammar is LL(1)

S→XS|dS|e

X→Y|Zb|aY

Y→cZ

Z→e

Q 3 (1*10=10)

Write Algorithm to construct SLR parser. Show all the steps of SLR parsing for following grammar.

S → R

R→ R b

R → a

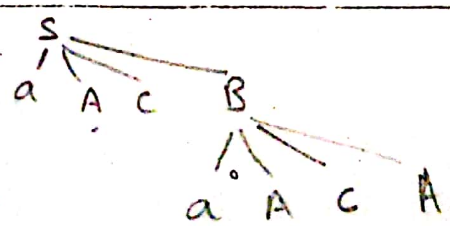
Q 4 Attempt both parts (2*5=10)

a) Draw the finite automata which can be used by lexical analyzer to recognize identifier, number and operators.

b) What is operator grammar? Write operator precedence parsing table for following grammar:

E→E+E|E*E|(E)|a

S→aAcB|Bds
B→aAcA|cAB|b
A→aB|aBc|a



Ms. Shalu

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aacbbcc

Scanned with CamScanner

Question 2(b) :-

$S \to XS \mid dS \mid \epsilon$

$X \to Y \mid Zb \mid aY$

$Y \to cZ$

$Z \to e$

Method 1:-

	First	Follow
S	a, c, e, d, ε	{$}
X	a, c, e	a, c, e, d, $
Y	c	a, c, e, d, $
Z	e	a, c, e, d, b, $

	a	b	c	d	e	$
S	S→XS		S→XS	S→dS	S→XS	S→ε
X	X→aY		X→Y		X→Zb	
Y			Y→cZ			
Z					Z→e	

No Multiple entries. Hence grammar is LL(1)

Method 2:-

For $S \to XS \mid dS \mid \epsilon$

$A_1 = First(XS) = First(X) = \{a, c, e\}$

$A_2 = First(dS) = \{d\}$

A_3 $First(\epsilon) = \{\epsilon\}$

$A_1 \cap A_2 \cap A_3 = \phi$ and only one Production has ε move.

For $X \to Y \mid Zb \mid aY$

$B_1 = First(Y) = \{c\}$ $B_1 \cap B_2 \cap B_3 = \phi$

$B_2 = First(Zb) = \{e\}$ and No ε-Move.

$B_3 = First(aY) = \{a\}$

For $Y \to cZ$ only a single Production

For $Z \to e$ only a single Production

Hence No rule of LL(1) is violated.

Hence the grammar is LL(1).

Question No. 3

1. $S \to R$
2. $R \to Rb$
3. $R \to a$

Follow(S) = {$}

Follow(R) = {b, $}

I_0:-
$S' \to .S$
$S \to .R$
$R \to .Rb$
$R \to .a$

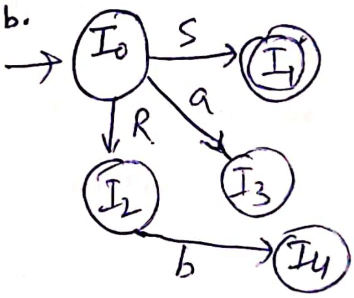
I_1:-
$S' \to S.$

I_2:-
$S \to R.$
$R \to R.b$

I_3:-
$R \to a.$

I_4:-
$R \to Rb.$

	a	b	$	S	R
I_0	S_3			1	2
I_1			accept		
I_2		S_4	R_1		
I_3		R_3	R_3		
I_4		R_2	R_2		
~~I_5~~					



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CSE-I

404- 24
405- 31
407- 10
65

(Please write your Roll Number) Roll No. ....

I P University
Second-Term Examination

SixthSemester (B.Tech) April-2015
Paper Code: ETCS-302 Subject: Compiler Design
Time: 1Hr 30 Mins Max. Marks: 30

Note: Q1 is compulsory andattempt any two questions out of remaining three.

Q1 Attempt any four parts (4*2.5=10)

a) What are the different representations of three address statements?
b) Give three address code for the following code segment :

While (a<b) do
  If(c<d) then x=y+z

c) Write syntax directed translation scheme for While loop.
d) Differentiate between SDT and SDD.
e) Write a brief note on responsibilities of semantic analysis phase.

Q2 Attempt both parts (2*5=10)

a) What is syntax directed translation? How are semantic actions attached to the production? Explain with an example.
b) Generate the intermediate code and output E.TRUE list and E.Falselist for the following expression:

x>y and z>k and not r<s from the following grammar
E→E or ME
E→Eand ME
E→ not E
E→ (E)
E→id
E→id relop id
M→ ϵ (marker non terminal)
Build up the parse tree.

Q 3Attempt both parts (5*2=10)

a) Explain L-attributed and S-attributed grammar with example.
b) Write syntax directed translation scheme for multi-dimensional arrays with suitable example.

Q4 Attempt both parts (2*5=10)

a) What is lexical phase, syntactic and semantic phase errors?
b) Discuss the various data structures used for symbol table with suitable example.

CSE-I

407 — 29

Roll Number.........................

First Term Examination

Seventh Semester [B.Tech.] (CSE/IT) Regular
September, 2015
Paper Code: ETCS-405
Subject: Compiler Construction
Time: 1.5 Hrs.
Maximum Marks: 30

Note: Question no. 1 is Compulsory. Attempt any more questions from the remaining.

Q1: (5 X 2=10)

(A) Name compiler construction tools.

(B) Define Token, Lexeme, and Pattern with suitable example.

(C) Difference between Kleen Closure and Positive Closure

(D) Write the regular expression for all the strings defined over alphabets (a, b) such that every string has at least one a and at least one b.

(E) What do you understand by the term sentential form and derivation tree of CFG.

Q2: For the regular expression **a.(a+b)*.b.bb** defined over input alphabets (a, b) do the following:

(A) Equivalent NFA with ε- move. (4)

(B) Conversion of derived NFA from part A to equivalent DFA. (4)

(C) Minimization of DFA derived from part B. (2)

Q3:

(A) Write note on Chomsky's Hierarchy of Grammars. (4)

(B) What is boot strapping? How it is used to generate cross compiler. (4)

(C) What are the various tasks done by the lexical analyzer? What is the significance of using 2-way input buffering in Lexical analyzer. (2)

Q4:

(A) What do you understand by Left recursion? How it can be removed? Explain with suitable example. (3)

(B) What do you understand by the ambiguity of a Grammar? How the ambiguity of Operator Grammar can be removed? (3)

(C) What do you understand by Top Down parsing? What are the main issues of it? Explain with suitable example. (4)

CSE-II ~~23~~ # 410 - 23

Roll Number.........................

Second Term Examination

November, 2015

Subject: Compiler Construction

Maximum Marks: 30

Seventh Semester [B.Tech.] (CSE/IT) Regular

Paper Code: ETCS-405

Time: 1.5 Hrs.

Note: Question no. 1 is Compulsory. Attempt any more questions from the remaining.

Q1: (5 X 2=10)

(A) What is S-attributed and L-attributed Grammars. Give suitable example for each.
(B) What is Peephole Optimization? Name some local code-improving transformations commonly used in Peephole Optimization.
(C) Give the structure of Activation Record and write the purpose of each segment.
(D) What do you understand by Back Patching? Explain with example.
(E) What are the various conflicts in LR Parsing.

Q2: Show that the following grammar is LR(1) but not LALR(1) (10)

S->Aa | bAc | Bc | bBa
A->d
B->d

Q3:

(A) Write the quadruple, triplet and indirect triplet for the expression -b*(c+d).
(B) Explain the Panic Mode Error Recovery and Phrase Level Error Recovery in LR Parser.
(C) Write note on Stack and Heap storage allocation strategies. (2,4,4)

Q4:

(A) For the Context Free Grammar { T->FT', T->*FT1' | ε , F->digit } do the following
(i) Write its equivalent L-attributed Grammar
(ii) Dependency Graph
(iii) Solution of the expression 3*5*2 through Annotated Parse Tree. (3,2,3)

(B) Find the basic block and flow graph for the sequence of following Three Address Code:

(1) PROD:=0
(2) I:=1
(3) T1:=4*I
(4) T2:= addr (A) – 4
(5) T3:= T2[T1]
(6) T4:= addr (B) – 4
(7) T5:= T4[T1]
(8) T6:= T3 * T5
(9) PROD:= PROD+T6
(10) I:= I+1
(11) IF I<20 goto (3) (2)

CSE - I 407 - 28

Roll Number........................

Second Term Examination

Seventh Semester [B.Tech.] (CSE/IT) Regular | **November, 2015**

Paper Code: ETCS-405 | **Subject: Compiler Construction**

Time: 1.5 Hrs. | **Maximum Marks: 30**

Note: Question no. 1 is Compulsory. Attempt any more questions from the remaining.

Q1: (5 X 2=10)

(A) What is S-attributed and L-attributed Grammars. Give suitable example for each.

(B) What is Peephole Optimization? Name some local code-improving transformations commonly used in Peephole Optimization.

(C) Give the structure of Activation Record and write the purpose of each segment.

(D) What do you understand by Back Patching? Explain with example.

(E) What are the various conflicts in LR Parsing.

Q2: Show that the following grammar is LR(1) but not LALR(1) (10)

S->Aa | bAc | Bc | bBa

A->d

B->d

Q3:

(A) Write the quadruple, triplet and indirect triplet for the expression -b*(c+d).

(B) Explain the Panic Mode Error Recovery and Phrase Level Error Recovery in LR Parser.

(C) Write note on Stack and Heap storage allocation strategies. (2,4,4)

Q4:

(A) For the Context Free Grammar { T->FT', T->*FT1' | ε , F->digit } do the following

(i) Write its equivalent L-attributed Grammar

(ii) Dependency Graph

(iii) Solution of the expression 3*5*2 through Annotated Parse Tree. (3,2,3)

(B) Find the basic block and flow graph for the sequence of following Three Address Code:

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(2) I:=1

(3) T1:=4*I

(4) T2:= addr (A) – 4

(5) T3:= T2[T1]

(6) T4:= addr (B) – 4

(7) T5:= T4[T1]

(8) T6:= T3 * T5

(9) PROD:= PROD+T6

(10) I:= I+1

(11) IF I<20 goto (3) (2)

CSE-I

101 = 19

19

(Please write your Roll no. immediately)

Roll No......................

First Term Examination

Seventh Semester [B.Tech.] (CSE/IT)
Paper Code: ETCS-405
Time: 1.5 Hrs.

September, 2013
Subject: Compiler Construction
Maximum Marks: 30

Note: Attempt Q.No. 1 which is compulsory and any two more questions from the remaining.

Q1 [5x2= 10]

a) Define token in reference to lexical analysis with suitable example.
b) Explain ambiguity of a CFG with example.
c) Define handle pruning.
d) What is a operator grammar? Check whether the grammar with productions $E \to EAE / id,\ A \to + / - / \epsilon$ is a operator grammar or not.
e) Give the left-most and right-most order of derivation for the string id+ id * id for a grammar with productions $E \to E+E,\ E \to E*E,\ E \to id.$

Q2

a) Find Leading and Trailing for the non-terminals of the grammar with productions $E \to E+T/T,\ T \to T*F/F,\ F \to (E)/id$ [4]

b) Construct operator precedence table for the grammar given above and check whether the String id+ id can be parsed for the grammar using operator precedence parsing. [6]

Q3

a) Compute First and Follow of all the non-terminals of the grammar given below $E \to TE',\ E' \to +TE'/\epsilon,\ T \to FT',\ T' \to *FT'/\epsilon,\ F \to (E)/id.$ [5]

b) Generate Predictive parsing table for the grammar specified above. [5]

First(E) = First of (T) = first of (F) = { (, id } First(E') = {+, ε}, First(T') = {*, ε}
Follow(E) = {$,)}, Follow(T) = {+, $,)}, Follow(E') = {$,)}, Follow(T') = {+, $,)}, Follow(F) = {*, +, $,)}

Q4 Write short notes on any two: [5x2=10]

a) LL(1) grammars.
b) Left recursion and left factoring.
c) Top down parsing and difficulties encountered in its implementation.
d) Chomsky classification.

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CSE-I, II

#507-5
#510-14
#510-14

(Please write your Roll no. immediately)

Roll No. 33

First Term Examination

Seventh Semester [B.Tech.] (CSE/IT)
Paper Code: ETCS-405
Time: 1.5 Hrs.

September, 2014
Subject: Compiler Construction
Maximum Marks: 30

Note: Attempt Q.No. 1 which is compulsory and any two more questions from the remaining.

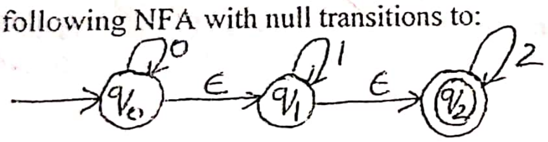
Q1 [5x2= 10]

a) Define token in reference to lexical analysis with suitable example.
b) Explain ambiguity of a CFG with example.
c) Define regular expression and regular sets.
d) What is a context free grammar? Give suitable example.
e) Give the left-most and right-most order of derivation for the string id+ id * id for a grammar with productions $E \to E+E / E*E / id$

Q2

a) Explain Chomsky classification of languages. [4]

b) Convert the following NFA with null transitions to:



i) NFA without null transitions
ii) DFA [6]

Q3

a) Compute First and Follow of all the non-terminals of the grammar given below
$E \to E+T/T$, $T \to T*F/F$, $F \to (E)/id$ [5]

b) Generate Predictive parsing table for the grammar specified above. [5]

Q4 Write short notes on any two: [5x2=10]

a) Check whether a grammar is LL(1) without construction of a parsing table.
b) Left recursion and left factoring.
c) Classification of parsing techniques.
d) Working of a LL(1) parser

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Ms Sakshi Hooda #308

IT

#405 - 03
#410 - 03
#407 - 04
10

Second Term Examination

Seventh Semester [B.Tech.]-Nov., 2014

Paper Code: ETCS-405	Subject: Compiler Construction

Time: 1.30Hrs. **Maximum Marks:** 30

Note: Q. No. 1 is compulsory and attempt any two more questions from the remaining.

Q.No. 1 (a) Define LR parsing and LR(K) Grammar.
(b) What is Annotated Parse Tree?
(c) Explain inherited and synthesized attributes.
(d) What is the Activation Record?
(e) Define Quadruples, indirect triples and triples.

[2 X 5=10]

Q.No. 2(a) Explain Errors occurring in different phases of compiler and Discuss Error recovery scheme in those phases. [10]

Q.No. 3(a) Construct SLR(1) parsing table for given Grammar:-
$E \rightarrow E+T$, $E \rightarrow T$, $T \rightarrow T*F$, $T \rightarrow F$, $F \rightarrow (E)$, $F \rightarrow id$ [6]

(b) Differentiate between LR(0), SLR(1), ClR(1) and LALR(1) parsers. [4]

Q. No. 4 (a) Explain static and dynamic memory allocation schemes with their merits and demerits. [5]
(b) Explain Code generation and its techniques. [5]

Ms Sakshi Hooda
#308

CSE-II

504 = 29

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(Please write your Roll no. immediately)

Roll No.....................

First Term Examination

Seventh Semester [B.Tech.] (CSE/IT)
Paper Code: ETCS-405
Time: 1.5 Hrs.

September, 2013
Subject: Compiler Construction
Maximum Marks: 30

Note: Attempt Q.No. 1 which is compulsory and any two more questions from the remaining.

Q1 [5x2= 10]

a) Define token in reference to lexical analysis with suitable example.
b) Explain ambiguity of a CFG with example.
c) Define handle pruning.
d) What is a operator grammar? Check whether the grammar with productions $E \rightarrow EAE / id$, $A \rightarrow + / - / \epsilon$ is a operator grammar or not.
e) Give the left-most and right-most order of derivation for the string id+ id * id for a grammar with productions $E \rightarrow E+E$, $E \rightarrow E*E$, $E \rightarrow id$.

Q2

a) Find Leading and Trailing for the non-terminals of the grammar with productions $E \rightarrow E+T/T$, $T \rightarrow T*F/F$, $F \rightarrow (E)/id$. [4]

b) Construct operator precedence table for the grammar given above and check whether the String id+ id can be parsed for the grammar using operator precedence parsing. [6]

Q3

a) Compute First and Follow of all the non-terminals of the grammar given below $E \rightarrow TE'$, $E' \rightarrow +TE'/\epsilon$, $T \rightarrow FT'$, $T' \rightarrow *FT'/\epsilon$, $F \rightarrow (E)/id$. [5]

b) Generate Predictive parsing table for the grammar specified above. [5]

Q4 Write short notes on any two: [5x2=10]

a) LL(1) grammars.
b) Left recursion and left factoring.
c) Top down parsing and difficulties encountered in its implementation.
d) Chomsky classification.

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Roll Number.........................

Seventh Semester [B.Tech.] (CSE/IT) Re-appear

Paper Code: ETCS-405

Time: 1.5 Hrs.

Second Term Examination

November, 2015

Subject: Compiler Construction

Maximum Marks: 30

Note: Question no. 1 is Compulsory. Attempt any more questions from the remaining.

Q1: (5 X 2=10)

(A) What is S-attributed and L-attributed Grammars. Give suitable example for each.

(B) What is Peephole Optimization? Name some local code-improving transformations commonly used in Peephole Optimization.

(C) Give the structure of Activation Record and write the purpose of each segment.

(D) What do you understand by Back Patching? Explain with example.

(E) What are the various conflicts in LR Parsing.

Q2:

(A) Find the Canonical Item set LR(1) for the following grammar

E->E+T | T

T->T*F | F

F->id | (E)

(B) Draw the Canonical Parsing Table for the above. (5,5)

Q3:

(A) Write the quadruple, triplet and indirect triplet for the expression -b*(c+d).

(B) Explain the Panic Mode Error Recovery and Phrase Level Error Recovery in LR Parser.

(C) Write note on Stack and Heap storage allocation strategies. (2,4,4)

Q4:

(A) For the Context Free Grammar { $T \to FT'$, $T' \to *FT_1' \mid \varepsilon$, $F \to digit$ } do the following

(i) Write its equivalent L-attributed Grammar

(ii) Dependency Graph

(iii) Solution of the expression 3*5 through Annotated Parse Tree. (3,2,3)

(B) Find the basic block and flow graph for the sequence of following Three Address Code:

(1) PROD:=0
(2) I:=1
(3) T1:=4*I
(4) T2:= addr (A) – 4
(5) T3:= T2[T1]
(6) T4:= addr (B) – 4
(7) T5:= T4[T1]
(8) T6:= T3 * T5
(9) PROD:= PROD+T6
(10) I:= I+1
(11) IF I<20 goto (3) (2)

(Please write your Roll no. immediately)

411 = 19
106 = 28
Roll No.......... 47

Second Term Examination

Seventh Semester [B.Tech.] (CSE/IT)
Paper Code: ETCS-405
Time: 1.5 Hrs.

October, 2013
Subject: Compiler Construction
Maximum Marks: 30

Note: Attempt Q.No. 1 which is compulsory and any two more questions from the remaining.

Q1 [5x2= 10]

a) Define LR parsing.
b) Explain LR(k) grammar with example.
c) Define synthesized translation and inherited translation with suitable example.
d) Explain block structured languages.
e) What is annotated parse tree?

Q2

LR(0) Parsing Follow

a) Construct the SLR parsing table for the given grammar (1) E→E+T, (2) E→T, (3) T→T*F, (4) T→F, (5) F→(E), (6) F→id [6]

b) Parse the string id*id+id using the parsing table constructed above. [4]

Q3

a) Explain the errors occurring in different phases of compiler and discuss the error recovery schemes in those phases. [10]

[5x2=10]

Q4 Write short notes on any two:

a) Various data structures used for symbol table management. → Linked list, Arrays, Hash Tables.
b) Different address codes in intermediate code generation. → Quadruples, Triples, Indirect Triple
c) Run time storage allocation and administration. → Activation. Rec ord.
d) Code generation and its techniques.

4→6

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Ms. Shalu

END TERM EXAMINATION

Exam Roll No. 1821520

SEVENTH SEMESTER [B.TECH] NOVEMBER-DECEMBER-2015

Paper Code: ETCS 405 | Subject: Compiler Construction

Time : 3 Hours | Maximum Marks :75

Note: Attempt any five questions including Q.No. 1 which is compulsory.

Q1. a) Discuss the need of a compiler. Discuss the basic structure of a compiler

b) Explain different types of code optimization techniques.

c) Discuss the advantage and disadvantage of creating Intermediate code.

d) Give short note on Context Free Grammar.

e) What is the role of Lexical analyzer phase in compiler? **(5x5=25)**

Q2. a) Write a brief note on classification of grammars. **(6)**

b) Find the reduced grammar which is equivalent to the CFG given below: **(6.5)**

S-> aC | SB
A-> bSCa
B-> aSB | bBC
C-> aBC | ad

Q3. a) What do you mean by parser? Explain its different types. **(6)**

b) Consider the grammar and test whether the grammar is LL(1) or not. **(6.5)**

S→ | AB | ε
A→ | AC | OC
B→ OS
C→ 1.

Q4. a) Explain the different data structure used for symbol table organization. **(6.5)**

b) What is an error? How does compiler handles various error in different phases? **(6)**

Q5. a) Explain the process of run time storage allocation. **(4)**

b) Explain various runtime storage management techniques? Explain with programming examples. **(8.5)**

Q6. Write short notes on:

a) Scanners **(4)**

b) Non DFA **(4)**

c) Polish N-tuple and syntax tree. **(4.5)**

Q7. a) How do we perform optimization with iterative loops? **(4.5)**

b) Write short note on Folding. **(4)**

c) How do we perform redundant sub-expression evaluation? **(4)**

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(Please write your Exam Roll No.) Exam Roll No. 05770103109

END TERM EXAMINATION

SEVENTH SEMESTER [B.TECH.] DECEMBER-2012

Paper Code: ETCS405 Subject: Compiler Construction

Time : 3 Hours Maximum Marks :75

Note: Attempt five questions including Q.no.1 which is compulsory. Select one question from each unit.

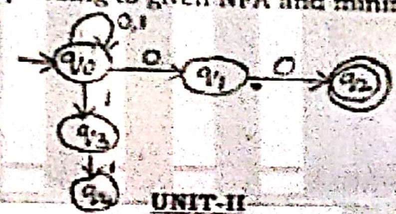
Q1 (a) Define the concept of bootstrapping of a compiler. (2)
(b) What is the difference between synthesized attributes and inherited attributes? (2)
(c) What do you mean by left recursion? How can it be removed? (3)
(d) What is an Operator Grammar? What are the constraints for a grammar to be operator grammar? (3)
(e) Differentiate between token, Lexeme and Pattern. (3)
(f) What is meant by Handle and Handle Prunning? (3)
(g) Define Backpatching with example. (3)
(h) What is peephole optimization? In what phase of compilation process, is it used? (3)
(i) Give the reverse polish notation for expression a+b*c/d+e. (3)

UNIT-I

Q2 (a) Give the Chomsky's classification of Grammar. (7.5)
(b) Draw the DFA for the regular expression 0*10*10*10*. (5)

OR

Q3 (a) What are the different phases of compiler? Explain each with example. (7.5)
(b) Genetic a DFA corresponding to given NFA and minimize it. (5)



UNIT-II

Q4 (a) Consider the context free Grammar G as $S \rightarrow AaAb$ $S \rightarrow Bb$ $A \rightarrow \epsilon$ $B \rightarrow \epsilon$. Find First and Follow sets. Is G LL(1)? Justify. (8)
(b) Draw the annotated Parse tree for the expression (3+4)*(5+6). Take necessary syntax directed definitions. (4.5)

OR

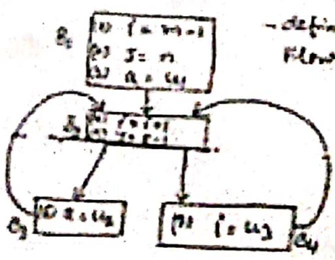
Q5 (a) Consider the following grammar $A \rightarrow AaAb$ $A \rightarrow BbBa$ $A \rightarrow \epsilon$ $B \rightarrow \epsilon$. Show whether Grammar is LR(1) or not. (7.5)
(b) What is symbol table? What are the various data structures used for symbol table implementation? (5)

UNIT-III

Q6 (a) Write notes on (i) Loop Unrolling and (ii) Loop Jamming. (5)
(b) Construct DAG for the following code:- (7.5)
(i) t1:=4*i (ii) t_2:=a[t_1] (iii) t_3:=4*i (iv) t_4:=b[t_3] (v) t_5:=t_2*t_4
(vi) t_6:=Prod+t_5 (vii) Prod:=t_6 (viii) t_7:=i+1 (ix) i:=t_7 (x) If i<=20 goto (1).

OR

Q7 Give the iterative solution for reaching definition of given flow graph. (12.5)



-definitions of given Flow graph.

UNIT-IV

Q8 What are the various techniques for allocating registers to variables in code generation phase? (12.5)

Q9 Describe the types of errors occurring in different phases of compiler. (12.5)

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